Experiment 16 - Experiment to understand EDA-Quantitative and Qualitative analysis.

Aim:

To understand EDA - Quantitative and Qualitative analysis.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

tips = sns.load_dataset('tips')
tips.head()

sns.displot(tips.total_bill, kde=True)
plt.show()

sns.displot(tips.total_bill, kde=False)
plt.show()

sns.jointplot(x=tips.tip, y=tips.total_bill)
plt.show()

sns.jointplot(x=tips.tip, y=tips.total_bill, kind="reg")
plt.show()

sns.jointplot(x=tips.tip, y=tips.total_bill, kind="hex")
plt.show()

sns.pairplot(tips)
plt.show()

tips.time.value_counts()
sns.pairplot(tips, hue='time')
plt.show()

sns.pairplot(tips, hue='day')
```

```
plt.show()

sns.heatmap(tips.corr(numeric_only=True), annot=True)
plt.show()
sns.boxplot(y=tips.total_bill)
plt.show()

sns.boxplot(y=tips.tip)
plt.show()

sns.countplot(y=tips.day)
plt.show()

sns.countplot(y=tips.sex)
plt.show()

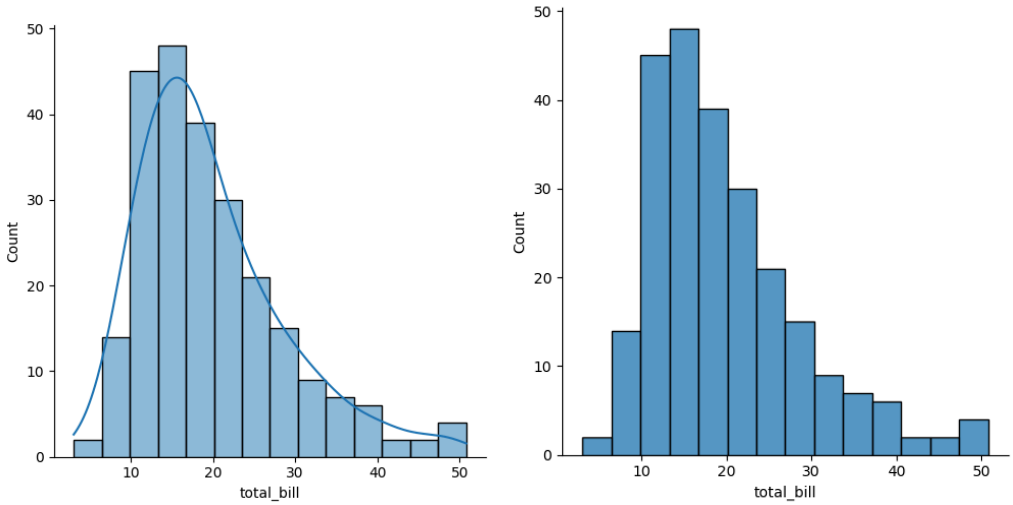
tips.sex.value_counts().plot(kind='pie')
plt.show()

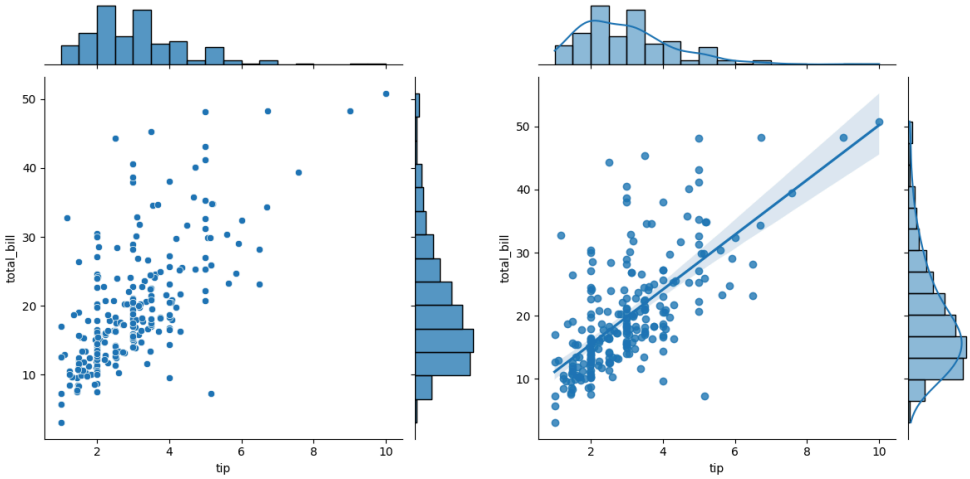
tips.sex.value_counts().plot(kind='bar')
plt.show()

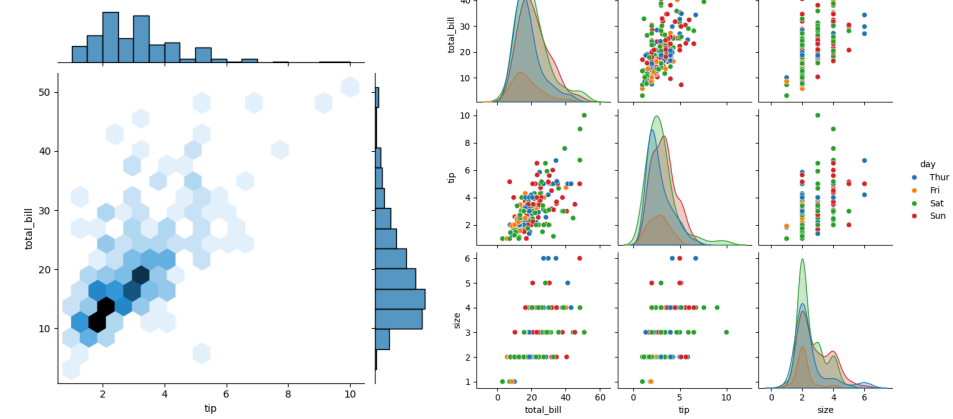
sns.countplot(y=tips[tips.time == 'Dinner']['day'])
plt.show()
```

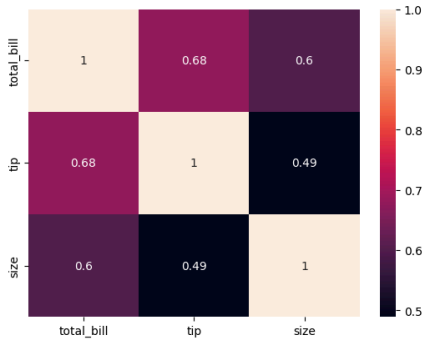
Output:

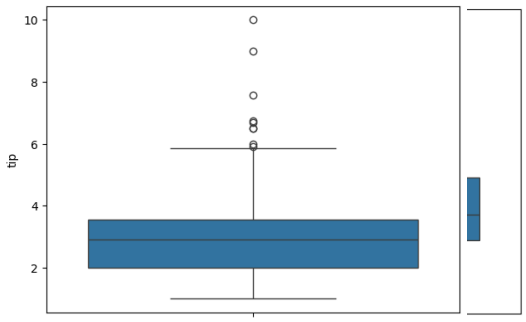
	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

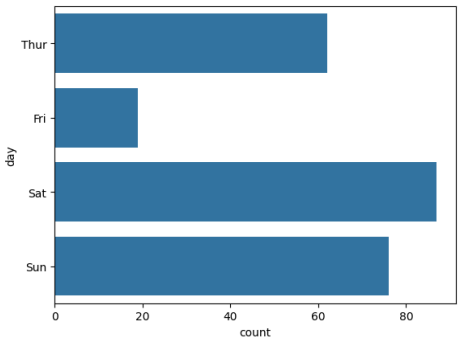

Count
total_bill

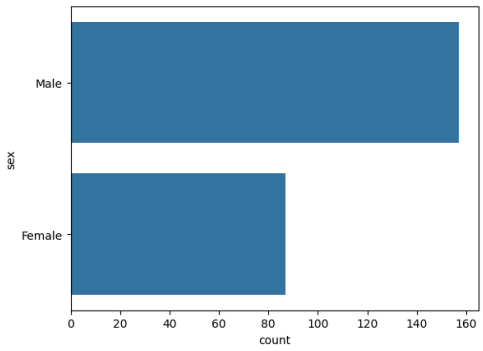

total_bill
tip

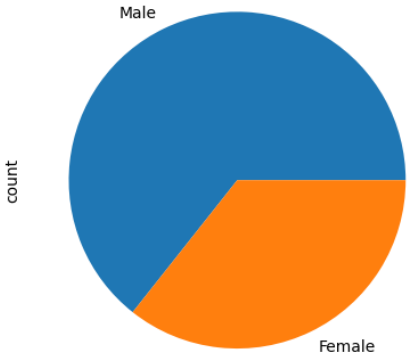

total_bill
tip
day
Thur
Fri
Sat
Sun
size

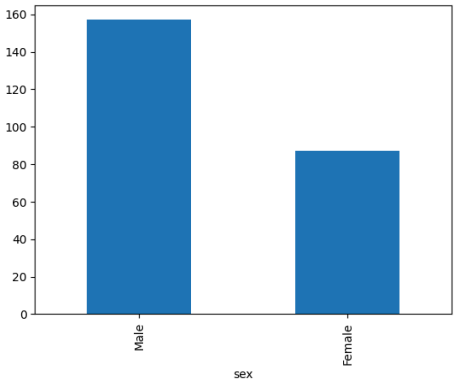

total_bill
tip
size
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
1.0
0.9
0.8
0.7
0.6
0.5

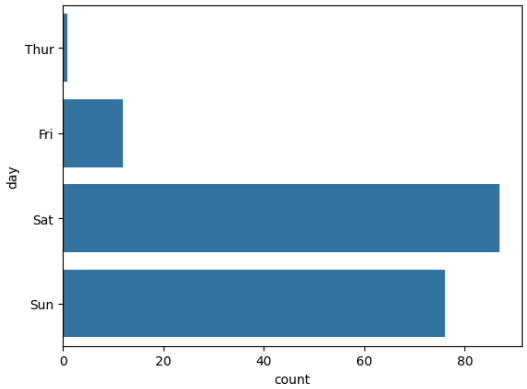

tip
10
8
6
4
2


day
Thur
Fri
Sat
Sun
0
20
40
60
80
count


sex
Male
Female
0
20
40
60
80
100
120
140
160
count


Male
count
Female





Result:

Thus, we have successfully understood EDA - Quantitative and Qualitative analysis.